

A MEAL'S ROAD MAP

The main regions of the digestive tract make up a tube 26-30 ft (8-9 m) long, and a meal passes through in 24-30 hours. The small intestine (p. 37), 21 ft (6.5 m) long, has been moved in this model to reveal the rest of the tract. Much of the digestive process is invisible to the naked eye, involving attacks on the food by natural chemicals called enzymes, which are made in the body. There are more than 20 specialized enzymes, each designed to digest one constituent in the food, such as a certain fat, carbohydrate, or protein.

MOVING ENERGY

All activities require energy, from driving a car to athlete Florence Griffith-Joyner winning a race.

The car's energy comes from chemicals in its fuel, gasoline. The body's fuel is food. Digestion converts starches and other carbohydrates into glucose, which is "burned" to release energy. Some of this is heat, and some is transformed in muscle cells into the energy of movement.



Left lung
(pp. 26-27)

Heart
(pp. 28-29)

Peripheral
nerves
in arm
(pp. 58-59)

Right lung

The liver acts as a
"chemical factory"

Gallbladder
stores bile fluid (p. 36)

The stomach is
a J-shaped bag
(pp. 36-37)

The large intestine
has been folded down

Food ends its
journey in
the rectum